

For Thread Handling

Design layout as:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ThreadActivity"
    android:orientation="vertical">

    <ProgressBar
        android:id="@+id/progressBar1"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:indeterminate="false"
        android:max="10"
        android:padding="4dip"></ProgressBar>

    <Button
        android:id="@+id/button1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="startProgress"
        android:text="Start" />

</LinearLayout>
```

Create Activity Class as:

```
package np.com.westernit.androidapp;

import android.os.Bundle;
import android.support.v7.app.AppCompatActivity;
import android.util.Log;
import android.view.View;
import android.widget.ProgressBar;
import android.widget.Toast;

public class ThreadActivity extends AppCompatActivity {
    private ProgressBar bar;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_thread);
        bar = findViewById(R.id.progressBar1);
    }

    public void startProgress(View view) {
        bar.setProgress(0);
        new Task().start();
    }

    class Task extends Thread {
        @Override
        public void run() {
            for (int i = 0; i <= 10; i++) {
```

```

        try {
            Thread.sleep(1000);
        } catch (InterruptedException e) {
        }
        bar.setProgress(i);
    }
    Log.e("AA", "Task Completed");
}
}
}

```

Register to Manifest File

```

<activity android:name=".ThreadActivity">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>

```

Handler Example

Create an Activity as:

```

package np.com.westernit.androidapp;

import android.app.ProgressDialog;
import android.os.Bundle;
import android.os.Handler;
import android.support.v7.app.AppCompatActivity;

public class HandlerActivityExample extends AppCompatActivity {
    private Handler mHandler;
    private ProgressDialog mProgressBar;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_handler_example);
        notificationDialog();
    }

    private void notificationDialog() {
        mHandler = new Handler();
        mProgressBar = new ProgressDialog(this);
        mProgressBar.setMax(100);
        mProgressBar.setProgressStyle(ProgressDialog.STYLE_HORIZONTAL);
        mProgressBar.show();
        new Thread(new Runnable() {
            @Override
            public void run() {
                for (int i = 0; i <= 100; i++) {
                    final int currentProgressCount = i;
                    try {
                        Thread.sleep(50);
                    } catch (InterruptedException e) {
                        e.printStackTrace();
                    }
                    //Update the value background thread to UI thread
                }
            }
        }).start();
    }
}

```

```

        mHandler.post(new Runnable() {
            @Override
            public void run() {
                mProgressBar.setProgress(currentProgressCount);
            }
        });
    }
}

}).start();
}
}

```

Design Blank Layout as

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".HandlerActivityExample">

</LinearLayout>

```

Register to manifest file

```

<activity android:name=".HandlerActivityExample">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>

```

AsyncTask Example

```

public class AsyncActivity extends Activity {

    private TextView tvResult;
    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_async);

        tvResult = findViewById(R.id.tvResult);
        new MyTask().execute();
    }

    private class MyTask extends AsyncTask<Void, String, Void> {

        @Override
        protected Void doInBackground(Void... voids) {

            for(int i=0;i<100;i++){
                publishProgress(""+i);
                SystemClock.sleep(100);
            }
        }
    }
}

```

```

        return null;
    }

    @Override
    protected void onPostExecute(Void aVoid) {

        tvResult.append("Complete Operation");
    }

    @Override
    protected void onPreExecute() {
        tvResult.setText("Task Start");
    }

    @Override
    protected void onProgressUpdate(String... text) {
        tvResult.append(text[0]);
    }
}

```

AsyncTask Example For ListView

```

public class AsyncActivity extends Activity {

    ListView lvResult;
    ArrayAdapter<String> adapter;
    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_async);

        lvResult = findViewById(R.id.lvResult);
        ArrayAdapter<String> adapter =
            new
ArrayAdapter<>(getApplicationContext(), android.R.layout.simple_list_item_1, new
ArrayList<String>());
        lvResult.setAdapter(adapter);
        new MyTask().execute();
    }

    class MyTask extends AsyncTask<Void, String, Void> {

        @Override
        protected void onPreExecute() {
            adapter = (ArrayAdapter<String>) lvResult.getAdapter();
        }

        @Override
        protected Void doInBackground(Void... voids) {
            for(int j=1001; j<10001; j++){
                publishProgress(j+"");
                SystemClock.sleep(100);
            }

            return null;
        }

        @Override
        protected void onProgressUpdate(String... values) {

```

```
        adapter.add(values[0]);
    }

    @Override
    protected void onPostExecute(Void aVoid) {
        Toast.makeText(getApplicationContext(), "All items are added  
Successfully", Toast.LENGTH_LONG).show();
    }
}
```